

CSC103-Programming Fundamentals

Selections

Lecture-7

Instructors

Mr. Rizwan Rashid

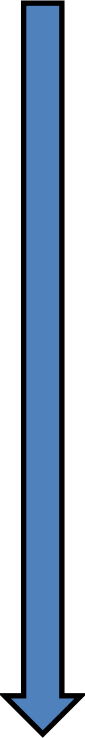
Lecture Outline

- Introduction
- Boolean Types, Values, and Expressions
- One-way **if** statement
- Two-way **if-else** statement

Introduction

- Control structure: logical design that controls order in which set of statements execute
- Sequence structure: set of statements that execute in the order they appear
- Decision structure: specific action(s) performed only if a condition exists
 - Also known as selection structure

Introduction



```
import java.util.Scanner;
public class ComputeArea{
    public static void main(String[] args){

        // Create a Scanner object
        Scanner input = new Scanner(System.in);
        // Prompt the user to enter a radius
        System.out.println("Enter the radius of Circle");

        double radius = input.nextDouble();

        // Compute area
        double area    = 3.14159 * radius * radius;

        // Display results
        System.out.print("The area of Circle is: "+area);
    }
```

Boolean Types, Values, and Expressions

- Selection statements use conditions, which are *Boolean expressions*.
- *A Boolean expression is an expression that evaluates to a Boolean value **True** or **False**.*
- *JAVA provides six comparison operators (also known as relational operators)*

Relational Operators

TABLE 3.1 Relational Operators

<i>Java Operator</i>	<i>Mathematics Symbol</i>	<i>Name</i>	<i>Example (radius is 5)</i>	<i>Result</i>
<	<	less than	<code>radius < 0</code>	<code>false</code>
<=	≤	less than or equal to	<code>radius <= 0</code>	<code>false</code>
>	>	greater than	<code>radius > 0</code>	<code>true</code>
>=	≥	greater than or equal to	<code>radius >= 0</code>	<code>true</code>
==	=	equal to	<code>radius == 0</code>	<code>false</code>
!=	≠	not equal to	<code>radius != 0</code>	<code>true</code>

boolean Data Type

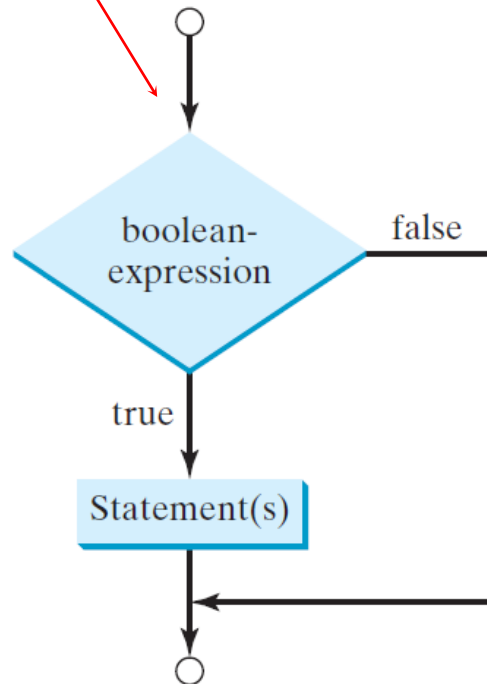
The `boolean` data type declares a variable with the value either `true` or `false`

- A variable that holds a Boolean value (`True/False`) is known as a **Boolean variable**.
- E.g. `boolean lightsOn = true`
- **true** and **false** are literals, like `10`, and are reserve words

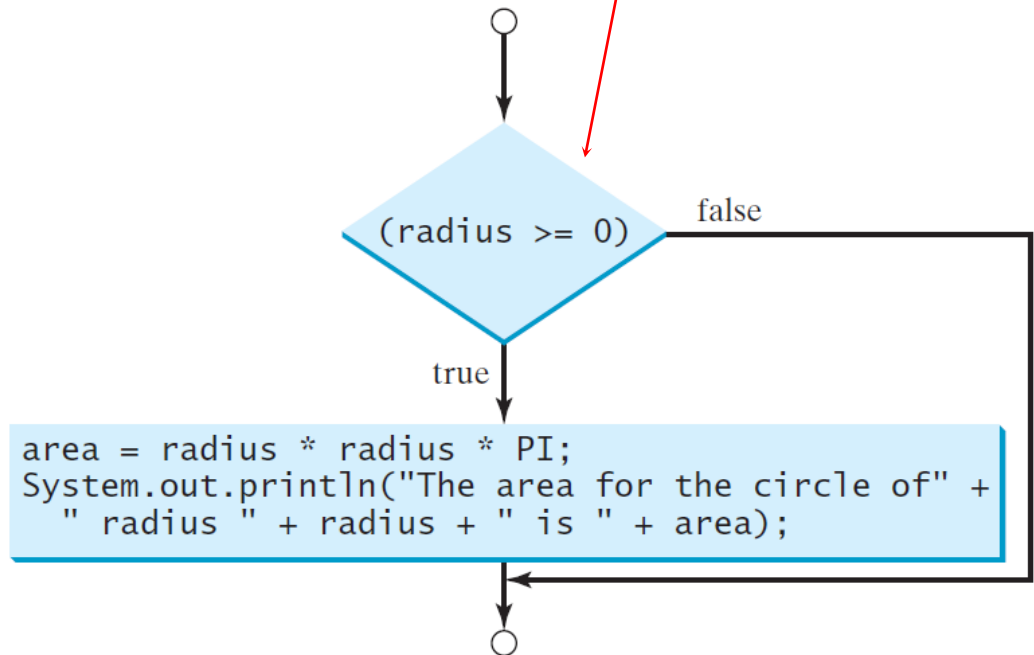
One-way if Statements

Syntax

```
if (boolean-expression) {  
    statement(s);  
}
```



```
if (radius >= 0) {  
    area = radius * radius * PI;  
    System.out.println("The area"  
        + " for the circle of radius "  
        + radius + " is " + area);  
}
```



Note

The boolean-expression is enclosed in parentheses. For example, the code in (a) is wrong. It should be corrected

```
if i > 0 {  
    System.out.println("i is positive");  
}
```

(a) Wrong

```
if (i > 0) {  
    System.out.println("i is positive");  
}
```

(b) Correct

The block braces can be omitted if they enclose a single statement. For example, the following statements are equivalent.

```
if (i > 0) {  
    System.out.println("i is positive");  
}
```

(a)

Equivalent

```
if (i > 0)  
    System.out.println("i is positive");
```

(b)

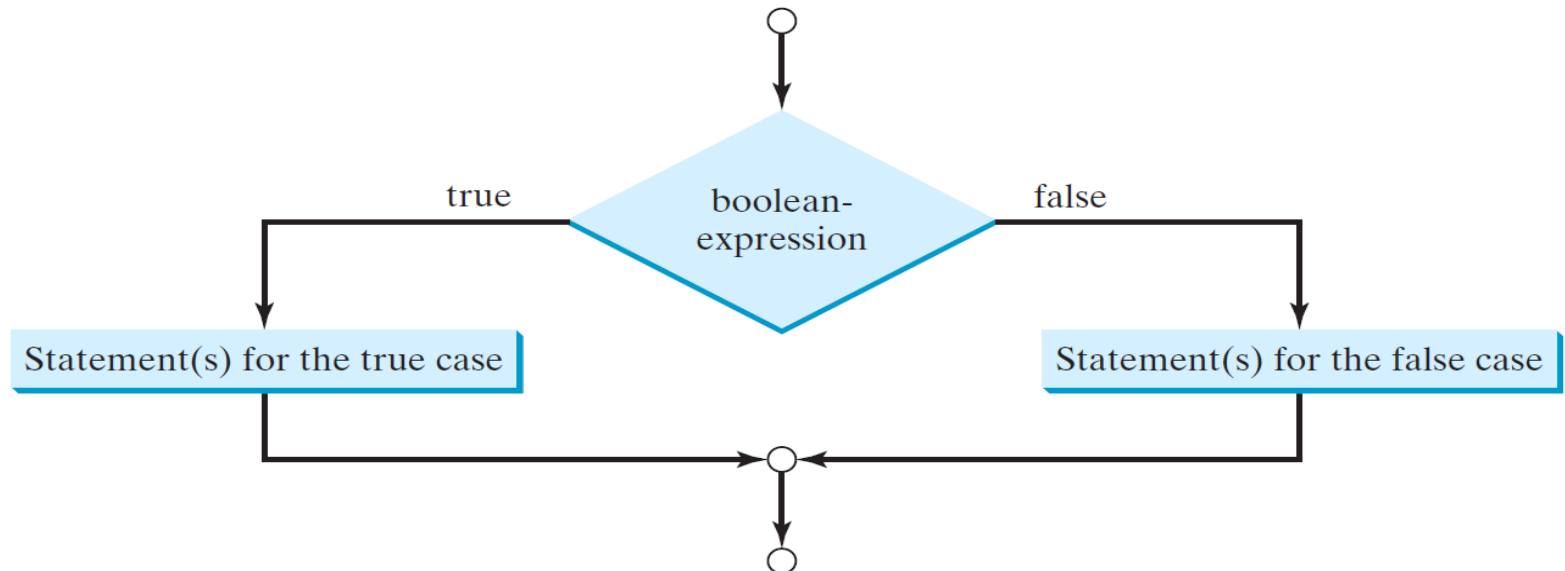
Problem

- Write a JAVA program that prompts the user to enter an integer. If the number is a multiple of 5, the program displays HiFive. If the number is divisible by 2, it displays HiEven

The Two-way if Statement

An if-else statement decides the execution path based on whether the condition is true or false

```
if (boolean-expression) {  
    statement(s)-for-the-true-case;  
}  
else {  
    statement(s)-for-the-false-case;  
}
```



if-else Example

```
if (radius >= 0) {  
    area = radius * radius * 3.14159;  
  
    System.out.println("The area for the "  
        + "circle of radius " + radius +  
        " is " + area);  
}  
else {  
    System.out.println("Negative input");  
}
```

```
int time = 2, tide = 3;  
if (time + tide > 6)  
    System.out.println("Time and tide wait for no one.");  
else  
    System.out.println("Time and tide wait for me.");
```

Write an if statement that increases pay by 3% if score is greater than 90

Write an if statement that assigns 1 to x if y is greater than 0.

Write an if statement that increases pay by 3% if score is greater than 90, otherwise increases pay by 1%.

Problem

Chris owns an auto repair business and has several employees. If an employee works over 40 hours in a week, Chris pays that employee 1.5 times his or her regular hourly pay rate for all hours over 40. Chris has asked you to design a simple payroll program that calculates an employee's gross pay, including any overtime wages.

Algorithm

1. Get the number of hours worked.
2. Get the hourly pay rate.
3. If the employee worked more than 40 hours, calculate the gross pay with over-time. Otherwise, calculate the gross pay as usual.
4. Display the gross pay.

Summary

- Boolean Expression
- Conditional Statements

Next Lecture

- Nested **if** statement
- Multi-way **if-else if-else** statement
- Logical Operators
- Conditional expressions